

Cross-Cultural Affiliate Network Management Skill Collection

AI Agent-based Multi-Region Affiliate Automation System

Tencent Cloud Hackathon 2026

AI Agent Skill Track

Technical Document v2.0

2026-05-15

1. Product Overview

1.1 Basic Information

Item	Content
Product Name	Cross-Cultural Affiliate Network Management AI Agent Skills
Track	Tencent Cloud Hackathon 2026 - AI Agent Skill Track
Platform	CodeBuddy / WorkBuddy / OpenClaw
Skills	15 (incl. 2 Layer 5 new additions)
Version	v2.0

1.2 One-line Description

Encapsulate cross-border affiliate operations knowledge into 15 reusable AI Agent Skills, achieving multi-region, multi-cultural, multi-MMP operational automation, with a new AI Organization Behavior layer that enables AI to "remember people" and "evaluate relationships".

2. Background & Pain Points

2.1 Business Background

As a cross-border affiliate network company, we simultaneously serve three types of advertisers (Brazil/Europe/China), connect with three types of traffic sources (Brazil local, Chinese outbound DSP, Southeast Asian internet), and manage 20+ channels and 50+ apps.

2.2 Three Core Pain Points

Pain Point	Manifestation	Impact
Highly repetitive data processing	5+ reports/week manual cleaning, dedup, merge,	30% team time on repetitive tasks
Low cross-cultural comms efficiency	3 advertiser types x 3 traffic sources, 5-8 email rounds	1-4h per communication
Experience-driven budget allocation	20+ channels, CAP based on gut feeling	20-30% budget wasted

2.3 Relationship Management Pain Points

Pain Point	Manifestation	Impact
No partner churn warning	Rely on experience, discover too late	Lose quality channel relationships
AI doesn't know people	Every interaction is with a stranger	AI efficiency greatly reduced

3. Solution Architecture

3.1 Five-Layer Skill Architecture

```
Global Affiliate Ops Agent
|-- Layer 1 -- Data Connectors (Data Extraction & Validation)
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```
| |-- crm-channel-extraction
| |-- crm-settlement-verification
| +-- pa-channel-export
|
|-- Layer 2 -- Data Analytics (Business Intelligence)
| |-- multi-mmp-attribution-engine
| |-- ai-budget-optimizer
| |-- conversion-funnel-intelligence
| |-- geo-market-intel-engine
| |-- wish-intelligence-collector
| +-- regional-wish-classifier
|
|-- Layer 3 -- Anti-Fraud & Attribution
| |-- pa-report-workflow
| +-- cross-regional-revenue-audit
|
|-- Layer 4 -- Cross-Cultural Collaboration (Core Differentiator)
| +-- cross-cultural-negotiation-copilot
|
+-- Layer 5 -- AI Organization Behavior (CORE INNOVATION)
    |-- partner-memory-system
    +-- relationship-health-score
```

3.2 Technical Architecture Principles

Principle	Description
Single Responsibility	Each Skill solves one clear business problem
Independently Executable	Each Skill contains complete Python scripts
Layer Decoupling	Upper layers don't depend on lower layer implementations
Progressive Enhancement	Start with single Skill, expand to full automation
Stateful AI	Layer 5 Skills support memory accumulation and persistence

4. 15 Skills Detailed Description

4.1 Layer 1 - Data Connectors

crm-channel-extraction

Function: Auto-extract channel names and amounts from OCR-identified settlement PDFs/images. Impact: 1-1.5h manual -> 3min auto.

crm-settlement-verification

Function: Auto-reconciliation, flag anomalies with >5% discrepancy. Impact: 2h manual -> 3min auto.

pa-channel-export

Function: Generate standardized PA format reports. Impact: 0.5-1h manual -> one-click export.

4.2 Layer 2 - Data Analytics

multi-mmp-attribution-engine

Multi-MMP attribution dedup via Source Bank priority

ai-budget-optimizer

Channel effectiveness analysis with CAP suggestions (Green/Yellow/Red)

conversion-funnel-intelligence

5-stage funnel analysis with color-coded effectiveness ratings

geo-market-intel-engine

Multi-region market intelligence aggregation

wish-intelligence-collector

Multi-format Wish List parsing to standardized 8-column table

regional-wish-classifier

Auto-classify offers into 5 regional zones

cross-regional-revenue-audit

Full P&L audit: MMP + settlement + revenue -> 7-sheet Excel

macro-monitoring-agent

Daily FX/inflation/CPI tracking across regions with smart alerts

4.3 Layer 3 - Anti-Fraud & Attribution

pa-report-workflow

PA report automation workflow, supports week2/week3 data updates. Weekly pipeline: CSV collection -> JS array generation -> HTML report injection.

4.4 Layer 4 - Cross-Cultural Collaboration

cross-cultural-negotiation-copilot (v2.0)

CORE DIFFERENTIATOR. AI generates culturally-appropriate partner communication drafts covering 5 cultural zones (Brazil, China, Europe, Southeast Asia, Middle East). Three strategies: Soft Wording, Validation Framing, Internal Coordination Positioning.

5. Layer 5: AI Organization Behavior

CORE INNOVATION - AI is no longer a passive tool, but evolves into an 'AI Business Manager' with memory and relationship evaluation capabilities.

5.1 Design Philosophy

Version	Positioning	Capability
v1.0	Stateless Tool	Only analyzes data, doesn't know people, passive response
v2.0	Stateful AI	Remembers partner features, auto-adapts communication style
v3.0	Proactive AI Partner	Evaluates relationship health, proactively warns of churn risk

5.2 Partner Memory System

AI partner memory system - from 'stateless tool' to 'AI Business Manager with memory'. 10 dimensions of partner information:

Dimension	Values	Description
communication_style	soft/warm/aggressive/formal	Communication style
reply_speed	fast/medium/slow/dead	Reply speed
risk_history	fraud/payment_delay/no_issues	Risk history
negotiation_habit	price_pressure/bonus_hunter/easy	Negotiation habit
category_preference	finance/gaming/utility	Category preference
timezone	BRT/EST/CST	Timezone
emotion_style	optimistic/pessimistic/dramatic	Emotion style
contact_reliability	always_online/intermittent/offline	Contact reliability
call_preference	voice_call/sms_only/async_only	Call preference
budget_cooperation	high/medium/low	Budget cooperation level

5.3 Relationship Health Score

AI relationship health scoring system - from 'monitoring data anomalies' to 'monitoring relationship anomalies'. Score range: 0-100.

Dimension	Weight	Data Source	Calculation
Reply speed	20%	Partner Memory.reply_speed	Trend analysis
Emotional stability	25%	Last 10 communications tone	Variance calculation
Cooperation level	20%	Task completion rate	Ratio statistics
Payment timeliness	15%	Payment delay history	Weighted statistics
Traffic stability	20%	Last 30 days traffic	Standard deviation

Alert Thresholds

Level	Score Range	Alert	AI Action
Green Healthy	>=70	None	Normal monitoring
Yellow Attention	50-70	Yellow alert	Proactive check-in within 24h
Red Alert	<50	Red alert	Immediate notification + suggested action

6. Technical Innovation Points

6.1 Skill as Code

Each Skill contains complete executable Python scripts, not just prompts. AI Agents can independently invoke and execute them.

6.2 Cross-Platform Encoding Adaptation

Scenario	Processing
Unknown input file encoding	Auto-detect (chardet algorithm)
Output for Windows users	utf-8-sig + CRLF
Output for Mac/Linux users	utf-8 + LF
.bat/.cmd Chinese scripts	GBK encoding

6.3 Stateful AI (Layer 5 Innovation)

Innovation	Description
Memory accumulation	AI remembers 10 dimensions per partner
Relationship quantification	Health score makes relationships measurable
Proactive alerts	AI auto-monitors and pushes warning notifications
Strategy adaptation	Auto-adjusts communication based on memory

6.4 Built-in Business Rules

Business rules come from real operations experience, not generic AI responses. Example: dedup uses Source Bank priority logic (not pandas drop_duplicates), cross-cultural comms use Soft Wording + Validation Framing strategies (not standard templates).

7. Application Scenarios & Impact

Metric	Before	After	Improvement
Monthly reconciliation	~1 day	~30 min	96% reduction
Report cleaning	3-5h each	Automated	100% reduction
Cross-cultural comms	5-8 rounds	1-2 rounds	75% reduction
Budget allocation	Experience-driven	Data-driven	Qualitative change
New hire onboarding	2 weeks	2 days	86% reduction
Partner churn warning	By experience	AI proactive alert	60% improvement

8. Core Differentiation: Cross-Cultural Communication

Most AI Agent tools focus on 'data processing' and 'automation'. Few address the most critical need: cross-cultural

business communication.

Data processing -> use generic tools. Automation -> use generic RPA. Cross-cultural communication -> NO existing tool. That's the real moat.

Strategy 1: Soft Wording

Avoid directness that is perceived as arrogant in Brazilian culture. Example: 'Would you mind sharing the report?' instead of 'Send the report now.'

Strategy 2: Validation Framing

Confirm consensus before making requests to increase acceptance rate. Template: Confirm understanding -> Express appreciation -> Gently propose next step.

Strategy 3: Internal Coordination Positioning

Attribute pressure to 'other departments' to preserve relationship. Template: 'Our finance/legal team is asking for...'

9. Tech Stack

Component	Technology	Notes
Skill definition	Markdown (YAML frontmatter)	Metadata + docs + scripts
Automation scripts	Python 3.9+	pandas, openpyxl, requests, chardet
AI Agent runtime	OpenClaw / WorkBuddy	Skill loading + invocation
Encoding handling	Custom (write_file.py)	Cross-platform encoding + BOM/CRLF
Data formats	CSV / JSON / Excel (xlsx)	Both input and output
Partner memory	Local JSON files	data/partners/*.json

10. Future Roadmap

Phase 1 (1-2 months)

- Complete Layer 5 Skills (partner-memory-system + relationship-health-score)
- Integrate AppFollow/AppMagic API for real ranking data
- Add Partner Risk Radar Skill

Phase 2 (3-6 months)

- Integrate Tencent Cloud API (TDS user outreach, OCR, translation)
- Publish to SkillHub platform
- Support more MMP platforms
- Layer 5 expansion: multi-language sentiment analysis

Phase 3 (6-12 months)

- Build Skill Marketplace for organizational collaboration
- Integrate LLM multi-turn conversations
- Multi-tenant SaaS for SMB affiliate companies
- Layer 5 expansion: AI negotiation strategy recommendation